SUPPORTING ADULT GUIDANCE

The perpendicular bisector

Constructions, Loci and Bearings · Lesson 3. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson introduces the perpendicular bisector both as a **construction** and as a **locus** — the set of all points equidistant from two fixed points. Learners first discover the locus by dragging a point and collecting equal-distance positions, then construct the line exactly with equal compass arcs, and finally apply it to find a midpoint and to see that the centre of any circle through A and B lies on it.

Driving Question: *Where are all the points that are the same distance from two others?*

The mathematics, in plain terms

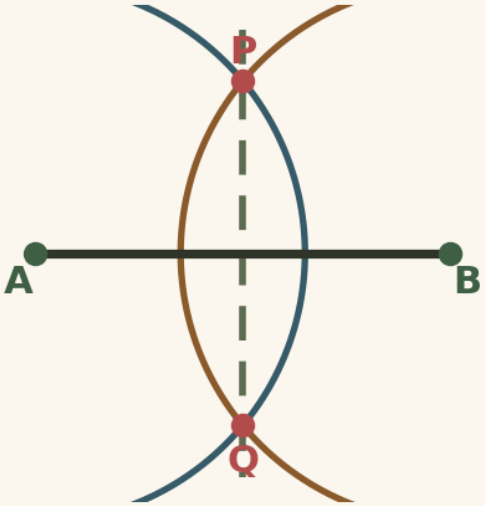
If a point is the same distance from A as from B, it is equidistant from them. The complete set of such points is a straight line: the perpendicular bisector of AB, which crosses AB at right angles through its midpoint. The construction works because equal compass arcs from A and from B meet at points that are, by construction, the same distance from both — two such points determine the line. A useful consequence: any circle passing through both A and B must have its centre equidistant from them, so its centre lies on the perpendicular bisector.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Construct the perpendicular bisector of AB

Construct the perpendicular bisector of the segment AB.

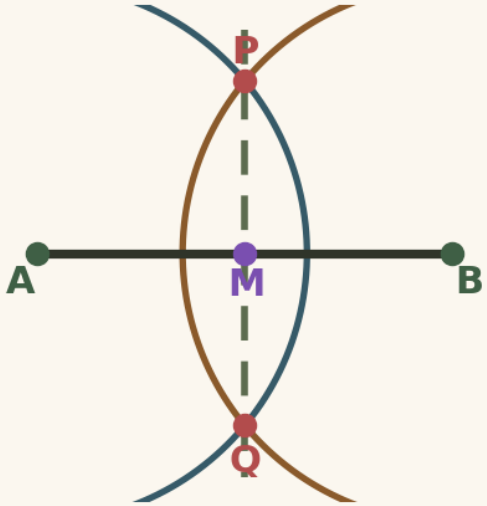


1. Open the compass wider than half of AB.	radius > ½ AB
2. Swing an equal arc from A (above and below).	arc from A
3. Keep the same opening; swing an arc from B.	arc from B
4. Mark the two crossings P and Q; draw the line PQ.	join P–Q

PQ is the perpendicular bisector: every point on it is the same distance from A and from B.

Example 2 — Use it to find the midpoint

The perpendicular bisector meets AB at its exact midpoint.



1. Construct the perpendicular bisector of AB (as above).	line PQ
2. Mark M where PQ crosses AB.	M on AB
3. M is the midpoint: MA = MB.	MA = MB

M is the exact midpoint of AB, found without measuring — MA = MB.

Locus first, then construction

It is tempting to jump straight to the compass recipe. The lesson deliberately lets learners *discover* the line first — by finding many equal-distance points and seeing them line up — before constructing it. Meeting the perpendicular bisector as 'all the points that…' plants the powerful idea of a locus, which the construction then makes exact. Encourage the learner to say what every point on the line has in common before naming it.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Find a fair meeting point equidistant from two homes, and notice such points line up.	~9 min
Movement	Drag a point to collect equal-distance positions; the perpendicular bisector emerges as their locus.	~13 min

Reflection	Construct the line exactly with equal arcs, and see why each crossing is equidistant from A and B.	~10 min
Creativity	Slide a centre along the bisector and watch its circle pass through both A and B.	~9 min
Rest	A short pause to let the widening of the question settle.	~2 min
Nutrition	Name what the learning feeds: the idea of a locus, and the pleasure of a fair, exact answer.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

“There is just one point the same distance from A and B (the midpoint).”

The midpoint is only one such point. Ask the learner to find another equal-distance point off the line AB — there are infinitely many, and they form a line.

“The arcs from A and B need different radii.”

For the perpendicular bisector both arcs use the same opening, wider than half of AB. Ask: 'If the crossing must be the same distance from A and B, what must the two radii be?'

“The bisector can be any line through the midpoint.”

It must be perpendicular to AB. Ask them to check the angle, or to test that points along their line really are equidistant from A and B.

Opening the compass narrower than half of AB.

Then the arcs never meet. Ask the learner to predict what happens and to widen the opening past the midpoint.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: equal-distance compass arcs (Lesson 1), and the distance between two points. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner has found the midpoint and thinks that is the whole answer, or has drawn a line through the midpoint that is not perpendicular. The aim is a question that hands the thinking back.

- “You found one point the same distance from A and B. Can you find another that is not on AB?”
- “What do all the equal-distance points you have found have in common?”
- “Both arcs used the same opening — so how far is each crossing from A, and from B?”
- “How could you convince someone your line really is the same distance from A and B all along it?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Do we need a protractor to get the right angle?

No. The equal-arc construction produces the right angle automatically; the perpendicularity is a consequence, not something you measure.

Why open the compass wider than half of AB?

If the opening is less than half, the two arcs cannot reach each other and never cross. Anything wider than half works.

Is 'locus' an important word?

Yes — it means the set of all points obeying a rule. The perpendicular bisector is the locus of points equidistant from two points, and loci recur throughout the unit.

What makes good evidence here?

A clear construction with arcs showing, the two crossings joined, and a sentence explaining that every point on the line is the same distance from A and B.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like

Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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